

IterETA

Personal Transportation Reliability and Vehicle Operations Platform

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Overview

IterETA is a personal transportation reliability and vehicle operations tool that enables users to better plan their commutes with more precise history-based estimated times of arrival (ETA). This utility will track performance and proactively monitor vehicle upkeep and safety risks. It is the result of the accumulated experience I, as a commuter student, have found the use for. Rather than directly handling navigation, IterETA prudently redirects users to existing map apps while focusing on services these apps may lack, including reliability analytics, risk alerts, and basic car ownership operations.

The proposed system aims to simplify decision-making by taking user trip, safety, and maintenance data and combining them into actionable insights via a web app interface and exportable CSV reports. To put it short, IterETA facilitates a more predictable and secure transportation experience.

Objectives

The main goals of this project are cleanly split into three.

1.) Store and organize trip, safety, and vehicle data

This project will focus on building a backend data system using Python and FastAPI, with a relational database implemented using SQLite and SQLAlchemy. Structured records of vehicles, trips, manually added safety incidents, and vehicle service history will be stored to support analytics features that enable planning and reporting.

2.) Use past trips to produce conservative arrival estimates and reliability metrics

Using data from previously recorded trips between repeated locations, travel time statistics will be generated. The system will also calculate conservative arrival estimates by adding a buffer for potential additions such as parking time, and will provide reliability measures, including travel time, delays, and on-time performance on repeated routes. All metrics will rely solely on locally stored data and will not use external data-providing services.

3.) Support safer and more predictable commutes using user-reported safety information and vehicle maintenance planning

Users will be able to enter safety incidents after trips using a user-assigned severity scale (1–10), and safety awareness alerts will be generated based on recent incident history and trip history. The system will also track vehicle maintenance and associated costs using the relational database and will create reminders for the user. These measures aim to improve commuting reliability and support safer travel by combining incident reporting and maintenance planning. CSV exports will also be available for further external analysis.

Overall Objective

React, Vite, TypeScript, and Tailwind will be used to create a web-based application that provides access to all system functionality.

Ultimately, this project aims to demonstrate reliable software engineering practices that encourage reproducible results and deliver a practical, well-scoped tool that remains within its intended bounds.

Deliverables

- A working IterETA web app for managing trip records, safety incidents, and vehicle data
- A backend system built with Python and FastAPI using a SQLite database with SQLAlchemy
- ETA estimation and commute reliability metrics sourced from stored trip data
- Manual safety incident logging with simple rule-based safety alerts
- Vehicle maintenance and cost tracking with service reminders
- A web interface built with React, Vite, TypeScript, and Tailwind
- CSV export of trip, safety, and maintenance data
- Database schema, migrations, backend tests, and setup documentation
- Screenshots and a short demo video of the completed system

Tentative Weekly Timeline

Week 1 Backend foundation

Set up the project, design the database schema, and implement the initial data models for vehicles, trips, safety incidents, and maintenance records.

Week 2 Backend foundation refinement and validation

Complete backend setup, refine the relational schema where needed, and validate basic create and query operations for all data entities.

Week 3 Trip and data workflows

Implement trip start and end functionality, safety incident entry, and vehicle maintenance and cost tracking.

Week 4 Trip and data workflows debugging and stabilization

Debug and stabilize trip workflows and data entry features, refine validation logic, and adjust the data model as necessary.

Week 5 ETA estimation and reliability metrics

Build the history-based ETA estimation and compute reliability metrics using stored trip data.

Week 6 ETA and analytics refinement and web interface integration

Refine ETA and reliability computations and begin integrating these features into the web application.

Week 7 Safety alerts and data export

Add simple rule-based safety alerts and CSV export for trip, safety, and maintenance data.

Week 8 Safety and export refinement and full system integration

Debug and refine safety alerts, export functionality, and complete integration of all backend features with the web application.

Week 9 Testing and system refinement

Add automated backend tests and perform system-wide debugging, refinement, and performance checks.

Week 10 Documentation and preparation

Write setup and usage documentation and prepare screenshots and a short demo video.

Week 11-12 Buffer Period

Take time to make crucial fixes, polish deliverables, and more importantly complete final report.

*** Note on Sources ***

IterETA is evaluated as a software system through functional correctness, usability, and performance of its backend and frontend components. The cited literature is used only to motivate the problem space and the relevance of the system.

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